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Pathway to Destruction: IL-17-Mediated ß-Cell Loss in Type 1 Diabetes

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Abstract

Type 1 Diabetes (T1D) is an autoimmune disease in which pancreatic ß-cells are destroyed, leading to little to no insulin being produced and impairing cellular glucose uptake. Over 8.4 million people worldwide are diagnosed with T1D, and there is currently no cure. Although it is established that pancreatic ß-cells are damaged in T1D, the molecular mechanisms behind this destruction is unclear. This research aims to explore the genetic factors contributing to T1D by analyzing gene expression profiles from blood samples of T1D patients and healthy controls from Gene Expression Omnibus (GEO). Differential gene expression analysis was conducted using GEO2R, and protein interaction networks were mapped using STRING and KEGG databases. The results showed 1,287 genes to be significantly dysregulated between T1D and control patients. Among the top 255 upregulated genes, there was significant enrichment of the interleukin (IL)-17 signaling pathway, which is heavily involved in inflammation and autoimmune response. These findings, corroborated by previous studies, suggest that the IL-17 signaling pathway may contribute to T1D by promoting inflammation that leads to ß-cell

apoptosis and therefore impaired insulin production.

Keywords: Type 1 Diabetes, IL-17 signaling pathway, differential gene expression, autoimmune response, pathway enrichment analysis

Introduction

An individual without any insulin would develop diabetic ketoacidosis within 1-2 days and, without treatment, would die within a week[1]. For the millions of people living with Type 1 Diabetes (T1D), this represents a persistent and very real threat. Type 1 Diabetes is a chronic condition in which the pancreas produces little to no insulin. Insulin is a hormone that allows glucose to enter cells, where it is used for energy in cellular processes. In T1D, significantly reduced insulin levels impair cellular function, leading to a range of serious symptoms and complications like unintended weight loss, fatigue, blurry vision, and high blood pressure.

T1D has a large national and global impact: globally, an estimated 8.4 million people have T1D[2]. In the US, 1.8 million Americans have T1D[3] and it was the eighth leading cause of death in 2021[4]. An estimated 64,000 people are diagnosed with T1D each year, and that number is expected to increase to 2.1 million by 2040[5], a rate four times faster than global population growth.

Beyond its physical symptoms, T1D imposes psychological burdens and significant difficulties in daily life. Current treatments require multiple insulin injections daily, constant glucose level monitoring, and a strict diet. With T1D, common activities like driving, exercise, working, and being pregnant become riskier. T1D patients are also more susceptible to other health issues like high blood pressure, heart problems, and high cholesterol[6]. Additionally, T1D patients and their families are also financially burdened by the diagnosis: the overall expense for diabetes patients is about 2.6 times greater than non-diabetic patients[7]. An estimated $16 billion is associated with T1D-associated healthcare expenditures and lost income each year[8]. Even if individuals are not personally diagnosed with T1D, they are likely to be impacted through friends and family who are.

Given the widespread and lasting consequences of T1D on patients and their families, continued research into its underlying causes is essential. It is known the immune system mediates destruction of pancreatic ß-cells, thereby impairing insulin production. A few genes, such as HLA-DRB1 and CTSH, and a positive family history have been shown to increase risk of T1D[9]. However, why these genes increase the risk and the mechanism behind this autoimmune process remain unclear. This paper investigates the genetic basis of the autoimmune response in Type 1 Diabetes in order to inform future treatments and diagnostic methods.

Methods

Gene Expression Omnibus (GEO) was used to find data on gene expression in patients without T1D (control group, n=20) and patients with T1D (n=20) in a study conducted by researchers at the University of San Paulo Hospital of Clinics. The dataset used was GSE193273: Prevalence of inflammatory pathways over immune-tolerance in peripheral blood mononuclear cells of recent-onset type 1 diabetes. This dataset contained 40 samples – 20 controls and 20 individuals recently diagnosed with T1D, when the immune system is most active[10]. Data was collected from peripheral blood mononuclear cells, which had previously been shown to reflect changes in innate and adaptive immunity occurring in and around pancreatic islets[10]. Gene expression for each patient was measured using the Agilent-072363 SurePrint G3 Human GE v3 8x60K Microarray[10].

Differences in gene expression between the control and T1D groups were analyzed using GEO2R, which calculated p-values and log fold change (logFC) values representing the relative difference in gene expression between the two groups. A negative logFC indicates a gene is upregulated in the T1D patients compared to the controls. GEO2R also created the figures in the results section, visualizing the difference in gene expression between the control and T1D groups. Lastly, GEO2R produced a table of differentially expressed genes that was imported into Google Sheets and sorted from lowest to highest logFC value.

From the Google Sheet, the top 255 genes with the most negative logFC values with a p-value less than 0.05 were imported into STRING-db, which then mapped the protein-protein interactions[11]. STRING-db identified several Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways associated with the differentially expressed genes. The IL-17 Signaling Pathway (KEGG ID: hsa04657) was chosen for further analysis because of its known role in autoimmunity and its statistical significance[12, 13].

Results

From GEO2R, we analyzed the differences in gene expression between the T1D and control groups and identified 1,287 genes that were significantly differentially expressed.

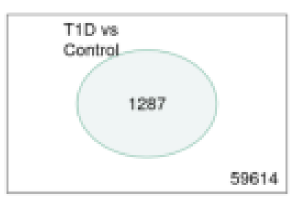

Figure 1: Venn Diagram of differentially expressed genes in T1D and Control patients[10]

GEO2R also created a volcano plot to visualize the differentially expressed genes. Each point represents a gene. The y-axis indicates the statistical significance of the expression differences, with a higher value indicating stronger significance. The x-axis shows the log fold change (logFC): positive logFC genes are downregulated in T1D and negative logFC genes are upregulated in T1D patients compared to control patients. As shown in the volcano plot, while many genes do not show significant differential expression, a substantial number are significantly dysregulated between T1D and control patients.

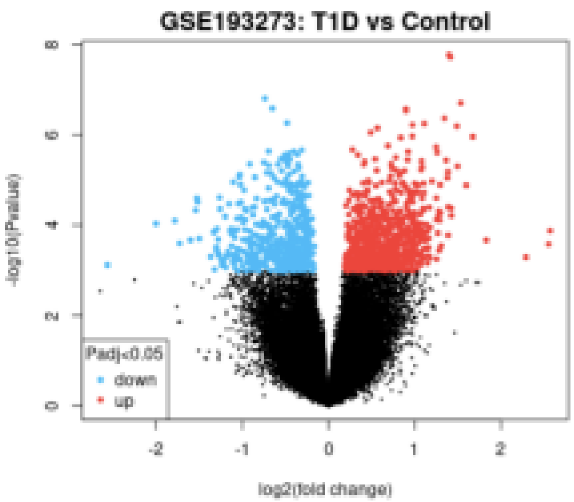

Figure 2: Volcano plot of differentially expressed genes in T1D compared to control patients[10]

GEO2R also generated a table of the top differentially expressed genes, as shown in Figure 3. The genes were sorted by logFC from lowest (upregulated) to highest (downregulated), with a p-value cutoff of 0.05. The most upregulated gene in our results was amphiregulin (AREG), with a logFC of -2.642 and p-value of 2.89E-03.

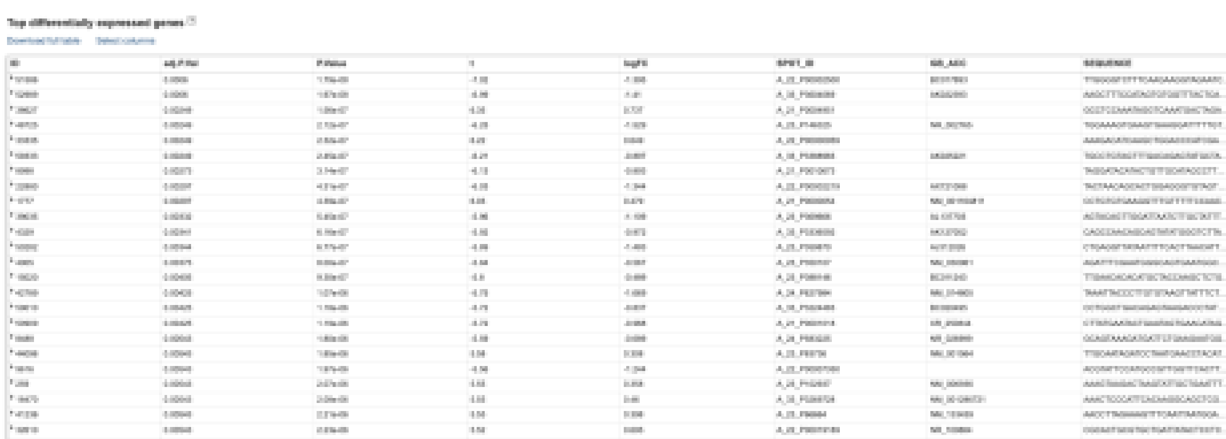

Figure 3: Table of top differentially expressed genes[10]

The top 255 upregulated genes were imported into STRING-db to map protein-protein interactions. Figure 4a shows the interactions between all inputted genes' products, while Figure 4b highlights dysregulated genes specifically involved in the IL-17 signaling pathway[11]. Each circle on the map represents a gene product dysregulated in T1D patients, and the connecting lines indicate interactions between them.

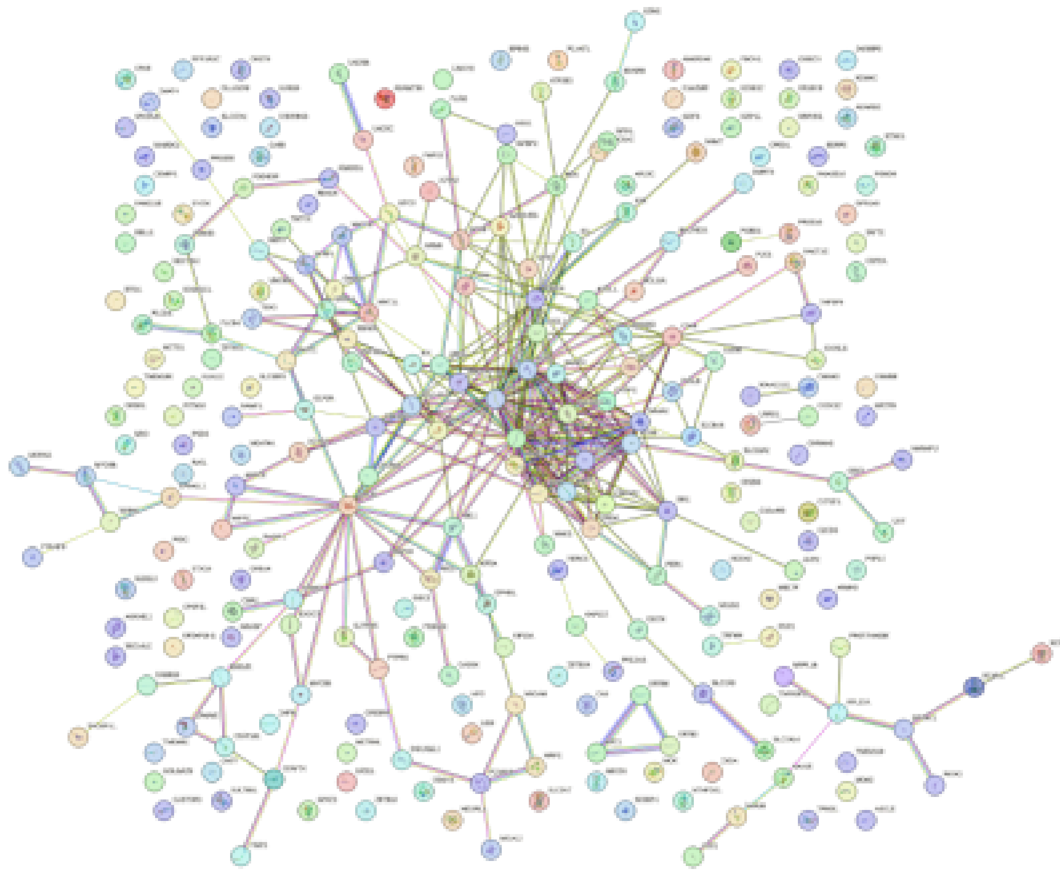

Figure 4a: STRING network of top 255 upregulated genes in T1D[11]

STRING-db identified several enriched biological pathways. The IL-17 signaling pathway (hsa04657) stood out due to its high enrichment score and relevance to autoimmunity[13]. The IL-17 signaling pathway is an inflammatory cytokine pathway that plays a role in host defense and immunopathology. Stimulated by the IL-17A cytokine, the IL-17 signaling pathway is known to increase inflammation, particularly in pancreatic cells, promoting ß-cell apoptosis.

Destruction and dysfunction of pancreatic beta cells reduces insulin production and impairs glucose regulation. This pathway was also statistically significant, with a count in network of 7 of 91, strength of 0.77, signal of 0.34, and false discovery rate of 0.0443[11].

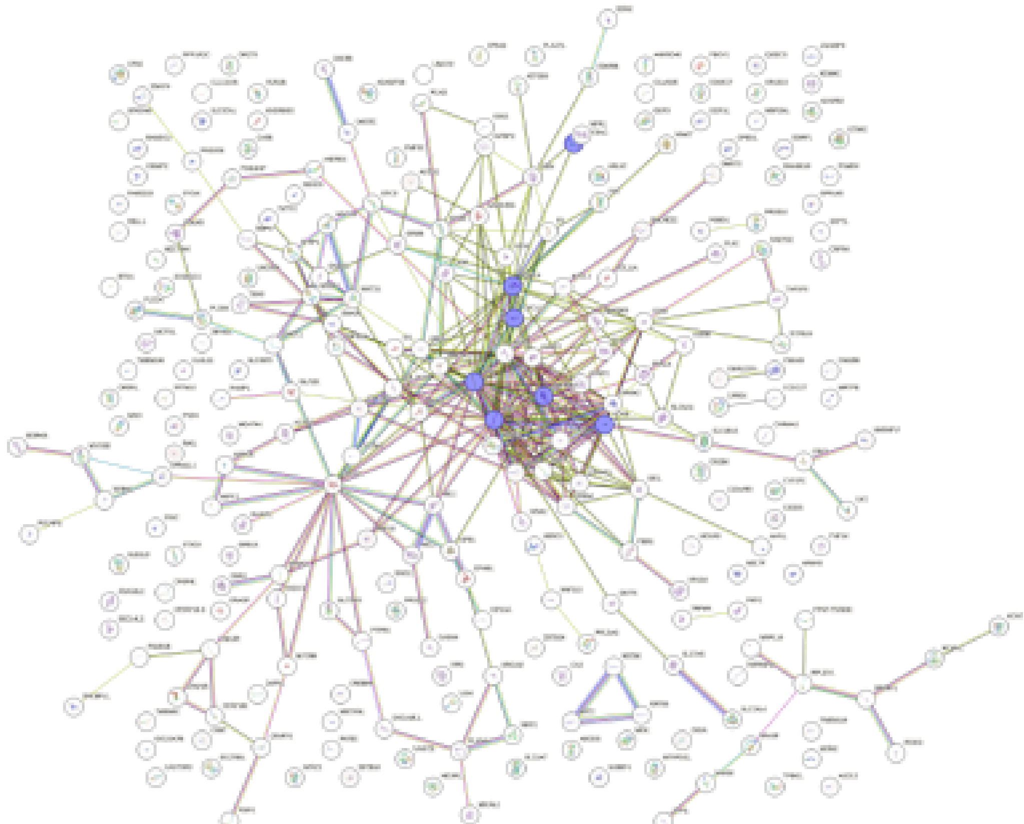

Figure 4b: STRING network of significantly upregulated proteins in T1D involved in the IL-17 signaling pathway[11]

Seven genes involved in the IL-17 signaling pathway were found to be significantly upregulated in T1D patients, consistent with the pathway's known role in promoting inflammation. The logFC value and p-value for these genes are shown in Table 1.

Gene	p-value	logFC
FOSB	1.99E-02	-1.120604
FOS	1.72E-03	-1.36937092
CXCL8	1.91E-02	-0.93785

JUN	1.60E-02	-0.97285611
CXCL1	5.17E-04	-0.83737905
TNFAIP3	6.04E-03	-0.92260304
MUC5AC	8.36E-03	-0.97045777

Table 1: logFC value and p-values of upregulated genes involved in the IL-17 Signaling Pathway
The IL-17 signaling pathway was further analyzed using the KEGG database, and the products of the upregulated genes in Table 1 were highlighted yellow in the pathway diagram.

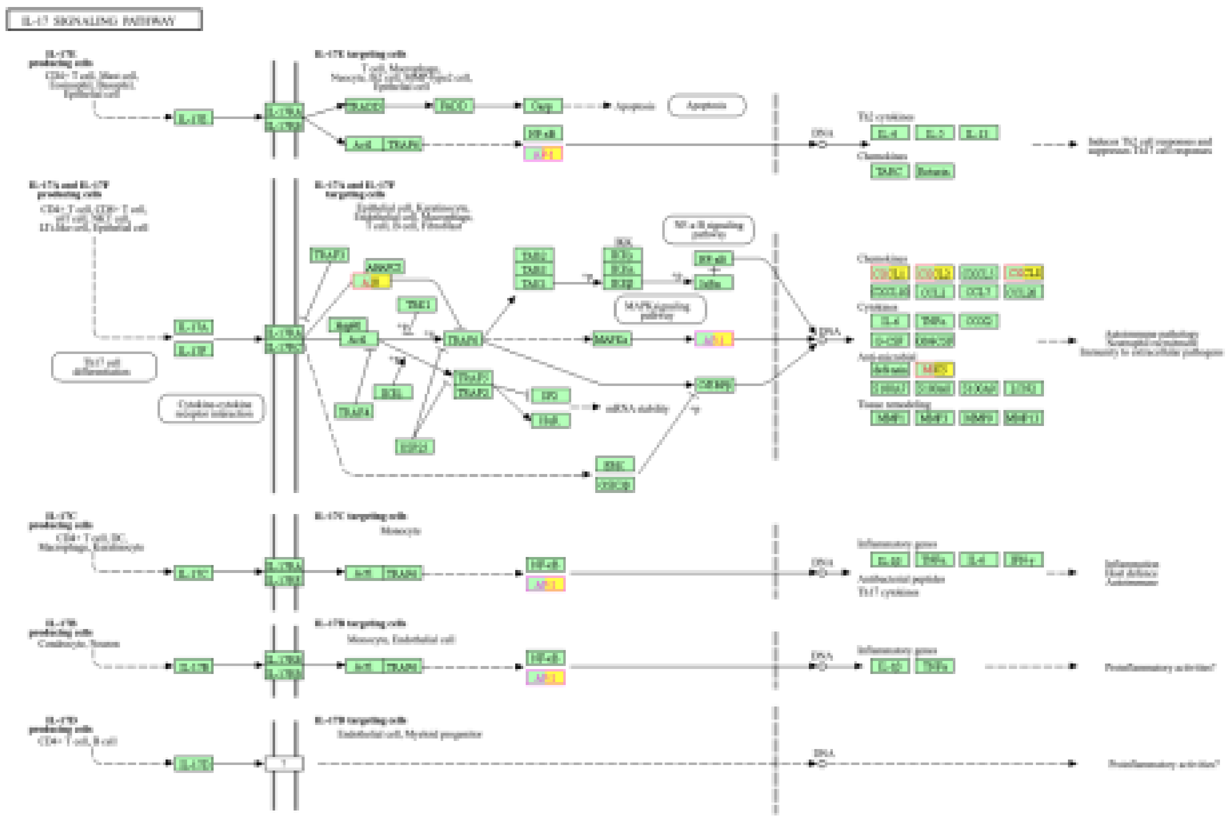

Figure 5: KEGG Pathway of IL-17 Signaling Pathway, with genes significantly upregulated in T1D highlighted yellow[12]

Many of the gene products upregulated in T1D are crucial to the autoimmune function of the IL-17 Signaling Pathway. For example, the protein AP-1, encoded for by FOS and JUN, plays an important role in recruiting neutrophils and inflammation. Chemokines such as CXCL1, CXCL2, and CXCL8 attract immune cells to their production site, in this case pancreatic cells. All of these genes were upregulated in T1D patients compared to control patients.

Discussion

Our transcriptomic analysis identified 1,287 genes differentially expressed between Type 1 Diabetes (T1D) patients and healthy controls. Among the top 255 upregulated genes, we observed a significant enrichment of the interleukin (IL)-17 signaling pathway, which plays an important role in autoimmune inflammation. The IL-17A cytokine stimulates inflammation, particularly within pancreatic tissues, which in turn attracts immune cells and promotes ß-cell apoptosis. As ß-cells are destroyed, insulin production declines, disrupting cells' ability to regulate blood glucose, the hallmark of T1D.

Seven genes—FOS, JUN, CXCL1, CXCL8, TNFAIP3, FOSB, and MUC5AC—were notably upregulated in T1D patients and involved in the IL-17 signaling pathway. Each of these genes contributes to the network of IL-17-mediated responses, which includes the recruitment of immune cells and the amplification of pro-inflammatory signals. The upregulation across these genes within the pathway underscores the pathway's over-active role in T1D and provides a potential explanation for its contribution to T1D pathogenesis.

For instance, CXCL1 and CXCL8, which are cytokines responsible for neutrophil recruitment, are upregulated in T1D. This suggests the IL-17 signaling pathway may be attracting abnormally greater levels of immune cells to pancreatic cells. In addition, FOS and JUN form the AP-1 transcription factor, which is involved in pro-inflammatory gene expression. These two genes are also upregulated, supporting the belief that T1D ß-cells are more prone to apoptosis due to increased inflammation. Interestingly, TNFAIP3, a negative regulator of NF-κB signaling, was also upregulated. This may reflect a feedback attempt to suppress excessive inflammation, though insufficient in halting disease progression.

Our results are consistent with previous studies linking IL-17 signaling to ß-cell damage.

Elevated levels of the cytokine IL-17A and increased activity of Th17 cells (immune cells that produce IL-17A) have been observed in T1D patients, often correlating with decreased β-cell function and elevated autoantibody levels[14]. Experimental studies further support this: in IL-17A-deficient Akita mice, researchers observed improved glycemic control, increased insulin levels, and lower expression of inflammatory cytokines compared to normal diabetic mice[15]. IL-17A has also been shown to directly impair human pancreatic islet function in vitro by amplifying inflammatory and apoptotic responses[14]. These findings are consistent with our hypothesis that increased expression of the IL-17 pathway contributes to ß-cell destruction and impaired insulin production. This research adds to current literature by explaining why this pathway is enriched in T1D patients: key genes in the IL-17 signaling pathway are upregulated in T1D patients, leading to increased expression of the IL-17 pathway.

Given its role, targeting the IL-17 pathway offers promising therapeutic potential. Monoclonal antibodies blocking IL-17A, such as secukinumab, and IL-17RA have already been used to successfully treat other autoimmune diseases and may be worth exploring for T1D. These therapies could potentially reduce ß-cell inflammation and preserve insulin production if introduced early in disease progression.

Our study has several limitations. We relied only on gene expression data without confirming corresponding changes at the protein level. Since changes at the RNA level do not always directly translate to protein abundance or function, future studies should use protein assays or functional experiments to assess the consequence of gene expression changes. Additionally, our dataset included only 40 samples, and the T1D group only included individuals recently diagnosed with T1D. As a result, the gene expression profiles may not fully represent individuals at later stages of the disease. These factors may limit the generalizability of our findings. Future studies should utilize larger and more diverse cohorts and study expression profiles throughout T1D onset and progression.

Conclusion

This study significantly strengthens the hypothesis that IL-17 signaling contributes to T1D pathogenesis by promoting inflammation, chemokine-driven immune-cell recruitment, and ß-cell apoptosis. The upregulation of several genes within this pathway highlights its crucial pro-inflammatory role in the autoimmune destruction of pancreatic ß-cells. These findings are consistent with existing experimental and clinical studies, and position the IL-17 pathway as a central intermediary in T1D pathogenesis. This supports the potential of IL-17 pathway inhibitors as therapeutic agents. This research not only deepens our understanding of the molecular mechanisms causing T1D but also highlights the IL-17 signaling pathway as a promising target for future therapeutic interventions aimed at preserving ß-cell survival and function. Future research should focus on validating the functional impact of these upregulated genes and exploring therapeutic strategies to regulate the IL-17 pathway.

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